

Pokemon Pricing App - Daily Update

Aug. 19, 2026 | *What's new since the last progress doc*

Short version: today was about turning the app from a price viewer into the start of an explainable buying tool. Price history is live, Recommended Buy has a first statistical baseline, comparable sales are much cleaner, and the liquidity work exposed the next data-collection problem.

1. Price history is live

- Added TCGplayer market-price history using TCGCSV's daily archives.
- History is backfilled offline and cached by set/group, so opening a card does not download archives.
- Chart supports 1M / 3M / 6M / 1Y / ALL; missing market-price days remain real gaps.
- Weekly sampling is the MVP default: enough for long-term movement without pulling roughly 3 GB of daily archives.
- Tested modern, vintage, fallback-priced and newly released cards; pre-release dates stay empty.

2. The main product number is now 'Recommended Buy'

The idea shifted from *"our estimated market price"* to: **"What price would actually look like a good buy right now?"**

- Stage A: Market Reference from TCGplayer + comparable eBay sales.
- Stage B: margin of safety based on spread, source disagreement, thin evidence, staleness and volatility.
- Median is the main eBay centre because the sale data is heavily right-skewed.
- The system can refuse a recommendation when the evidence is messy instead of forcing a fake-precise number.

Example output

TCGplayer Market	\$600
Recent eBay Median	\$585
Recommended Buy	<= \$560

3. Comparable sales got much cleaner

The pricing analysis found that the same card number can still contain totally different markets, so the grouping model was tightened.

- Added edition: 1st Edition / Shadowless / Unlimited / Other / Unknown.
- Added print variant: Holo / Reverse Holo / Non-Holo / Other / Unknown.
- Signed, sketched, altered and aftermarket art-card listings are excluded from normal comps.
- Card-number matching checks the denominator too, stopping Base Set 2 from contaminating Base Set groups.
- Language is now part of the group identity: explicit foreign sales stay visible but do not enter the English pricing group.
- A narrow Gold Foil Card rule caught another novelty Shining Charizard without rejecting legitimate Gold Secret Rares.

Result: the holdout set went from 3 false-positive comps to 0, while the development evaluation stayed unchanged.

4. The eBay UI now matches the analysis

- The old UI effectively put every raw sale into one bucket.
- Now it uses the exact same comparableGroup key as the Python analysis.
- Users can switch between markets like Raw - 1st Edition, Raw - Reverse Holo, PSA 10, PSA 9, etc.
- Each group has its own comparable-sale count.
- A parity script verifies that the English listings shown in the UI are exactly the listings used by Python.

5. Liquidity research found the real bottleneck

The '62-day dataset' was not actually a 62-day observation window. Every fixture is the 20 most recent sales for that card.

- Fast cards can fill 20 sales in 1-4 days; slow cards can take 50-60+ days.
- So fast markets are count-censored: we know they are liquid, but the true rate is above what we can observe.
- Days-since-last-sale is biased toward zero because each scrape starts from the newest sales.
- Only 5 of 144 comparable groups currently support even a prototype rate estimate.
- A 0-100 score was tested and rejected for now because it creates fake precision.

Best liquidity display later: show the evidence. With enough data: 'Typical comparable sale every ~3 days' + recent sale count. With sparse data: '3 comparable sales observed' + last sale date + an insufficient-evidence note.

Where things stand

- Price history: working
- Recommended Buy methodology: first explainable prototype complete
- Comparable groups: much cleaner and production/Python parity-checked
- Liquidity methodology: designed, but the collection method is now the bottleneck
- LLM fallback: still intentionally off in production

Next move

Redesign eBay history collection around a fixed time window. Instead of 'give me the 20 newest sales,' collect the same window (for example 90 days), save sales persistently, deduplicate by itemId, and record when a result is truncated.

Why: once every card is observed over the same window, we can calibrate a real green/yellow/red liquidity indicator and decide how liquidity should affect Recommended Buy.